Multi Dimensional Array(2-D Array)

Address Calculation in a multi Dimensional Array

1. **An array Arr[15][20] is stored in the memory along the row with each element occupying 4 bytes of memory. Find out the Base address and address of the element Arr[3][2], if the element Arr[10][25] is stored at the address 1500.**

Solution:

Total no. of Rows R=15

Total no. of Columns C=20

Lowest Row lr=0

Lowest Column lc=0

Size of element W=4 bytes

Arr[I][J] i.e., Arr[5][2]=1500

Arragement Order:Row wise

Base Address B=?

Arr[I][J]=B+W(C(I-lr)+(J-lc))

Arr[5][2]=B+4(20(5-0)+(2-0))

1500=B+408

B=1092

Base Address=1092.

Arr[3][2]=B+W(C(3-0)+(2-0))

=1092+4(20(3-0)+(2-0))

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=1092+248

=1340

Arr[3][2]=1340.

2. **An array VAL[1..15][1..10] is stored I the memory with each element requiring 4 bytes of storage. If the base address of array VAL is 1500, determine the location of VAL[12][9] when the array VAL is stored (i) Row wise (ii) Column wise**

Solution:

Base address B=1500

Element width w=4 bytes

Total rows r=15

Total columns c=10

ARR[I][J] = ARR[12][9] => I=12, J=9

Lowest row index lr=

Lowest column index lc=0

(i) **Row wise**

VAL[I][J] = B + w(c(I – lr) + (J – Lc))

VAL[12][9] = 1500 + 4(10(12-1) + (9-1))

= 1500 + 472

= 1972

(ii) **Column wise**

VAL[I][J]=B+W(I-lr)+R(J-lc)

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$=1500+4((12-1)+15(9-1))$

$=1500+4(131)$

$=1500+524$

$=2024$

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